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Highlights

- We incorporate reference price effects into a deteriorating inventory problem.
- The demand rate depends on displayed stock level and selling price simultaneously.
- The optimal pricing and inventory strategies are determined for each demand behavior scenario.
- The initial reference price plays an important role in making pricing strategies.

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Optimal dynamic pricing for deteriorating items with reference price effects when inventories stimulate demand

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Abstract

In this paper, we incorporate reference price effects into a deteriorating inventory problem when the demand rate depends on displayed stock level and selling price simultaneously. Reference price effects are formed empirically by customers to adjust current price perception on the basis of past purchases. The market demand is assumed to be responsive to the difference between the current selling price and the reference price, which is perceived as a loss or a gain. We consider three scenarios of market demand: loss neutrality, loss aversion, and loss seeking. An optimal dynamic pricing model is established to determine a pricing strategy maximizing the discounted total profit over an infinite time horizon. Theoretical results are derived to demonstrate the existence of the optimal solutions and to explore the relationships between optimal pricing decisions and an initial reference price based on mild assumptions. Numerical example and sensitivity analysis are presented to illustrate the theoretical results and managerial insights. Finally, concluding remarks are offered.

Keyword: inventory, pricing, stock-dependent demand, deterioration, reference price

1. Introduction

The importance of reference prices to a consumer has been repeatedly emphasized by both academicians and marketers. As is well known, customers are supposed as a rational agent who makes decisions based on current prices, income, and market conditions in traditional operational models (Popescu and Wu (2007)). Besides external market conditions, many marketing academicians and practitioners have recognized the importance of reference prices to customers' purchase decisions. Reference prices are most commonly defined as internally held standards that is formed based on past price exposures and used by customers to evaluate new pricing information. Kalyanaram and Winer (1995) provided empirical evidence to support that customers' purchase decisions are influenced by the reference price to a large extent. In a market with repeated interactions, consumers develop price expectations or their own ideas of a "fair price" referred to as the reference price from the past observed prices of the product. If the current selling price is lower than the reference price, it would result in a perceived gain and customers are more attracted to make the purchase.

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The opposite situation would result in a perceived loss, and customers would be less inclined to make the purchase. If the consumers are more responsive to a gain than a loss, they are called loss-seeking; otherwise they are loss-averse. Customers are loss-neutral if their response to both is the same.

However, most existing models in this stream of literature of joint pricing with reference price effects fail to take into account the demand-stimulating role of retail inventories. In contrast to constant or stochastic demand rates, for certain items, the displayed stock level has a positive impact on sales and profits. This happens because large amounts of inventory are more visible than small ones, and this increased visibility stimulates demand by creating cross-selling and impulse-buy opportunities. Wolfe (1968) found empirical evidence that sales of merchandise, such as women's dresses or sports clothes, are roughly proportional to the displayed inventory. Lusch and Dunne (1990) observed that high-performance retailers such as Toys "R" Us, Wal-Mart and Hypermart USA display enormous quantities of merchandise because product visibility might kindle latent demand. Other marketing researchers, for example, Corstjens and Doyle (1981), Desmet and Renaudin (1998) and Koschat (2008), also recognized that retail inventories serve a promotional role by stimulating demand. To quantify the phenomenon that inventories stimulate demand, inventory theorists have started formulating models by considering demand rate to vary with inventory level.

In the grocery retailing industry, mass merchandisers and warehouse clubs have become low price leaders by buying in enormous volume, but grocery stores operate with extremely low margins and depend on volume to generate profits (First Research (2013)). Poor pricing and inventory decisions can easily damage the profitability of deteriorating items in a retailer. Furthermore, there is compelling empirical evidence that the reference price and the amount of displayed inventory affect demand, but little researchers pay attention on these effects simultaneously. In this paper, we incorporate the reference price effect into an inventory model of dynamic pricing for deteriorating items when demand rate depends on both inventory level and selling price. All sale revenues and relative inventory costs are expressed in beginning-of-period cash units whereby cash flows occurring in subsequent time periods are discounted again by a one-period discount factor. We aim to investigate how stock level and consumers' price expectations reflect in managerial pricing decisions when customers have repetitive long-term relationships with the retailer. We also investigate the interactions among pricing, deteriorating rate, shape parameter of stock level and the reference price, and analyze how the initial reference price affects retailer's optimal pricing strategy. This paper contributes to two streams of literature: (1) the infinite planning horizon inventory model for deteriorating items with price- and stock-dependent demand, and (2) dynamic pricing with reference price effects. Whereas the first stream of literature assumes that the ending inventory levels are restricted to zero and that demand depends only on the stock level and current price, the second stream typically ignores the effect of stock level on demand and the deterioration of products.

The remainder of the paper is organized as follows. In Section 2, we review the literature on stock-dependent demand and reference price, respectively. In Section 3, we first describe the assumptions and notation used throughout this study. We then establish the discounted total profit function to the deteriorating inventory model with reference price effects, and demonstrate the existence of an optimal solution to the inventory problem in different customer behavior scenarios. Furthermore, several structural properties of optimal pricing and inventory strategies for each demand behavior scenario, such as the convergence behavior and the relationships between the initial reference price and optimal pricing strategy, are investigated to aid understanding the problem. In Section 4, some

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numerical examples and the results of a sensitivity analysis are discussed to illustrate the features of the proposed model. Finally, the conclusions and suggestions for future research are given in Section 5. The proofs are provided in the online Supplementary Material Part A.

2. Literature review

The primary research streams underlying our work are studies of dynamic pricing and inventory management with reference price effects when inventories stimulate demand. The relevant work and knowledge are reviewed below to clarify our contribution.

Reference prices are most commonly defined as internally held standards that consumers use to evaluate new pricing information. Regardless of how a reference price is formulated, actual prices below the reference price are usually perceived as fair, and those above it are viewed as unfair. Kalyanaram and Winer (1995) were the first to study the effect of reference prices on customer brand choices, and found that reference prices exerted a consistent and substantial influence on consumer decision making. Greenleaf (1995) indicated that reference price effects exert a positive impact on promotion profits and determined the optimal strategy of recurring promotions maximizing profits from such effects. Kopalle and Winer (1996) generalized the results of Greenleaf (1995) to reference price and product quality and demonstrated that the optimal strategy for loss-seeking customers is to institute a cyclical (High-Low) pricing policy. Fibich et al. (2003) studied dynamic pricing under deterministic linear demand in a continuous time framework when the reference price is an exponentially weighted average of past prices. They found that the steady-state price exists only when consumers are loss-neutral or loss-averse and indicated that a firm should adopt a skimming or a penetration strategy depending on the initial reference price. Popescu and Wu (2007) extended these results in Fibich et al. (2003) to general demand functions when the time is discrete. Nasiry and Popescu (2011) considered a similar deterministic demand model in which the reference price mechanism is based on a weighted average of the lowest price observed in the past and the most recent price. Hu et al. (2016) analyzed a dynamic pricing problem in a market facing reference price effects at an aggregate demand level when gains have greater impact than losses. Under relatively strict assumptions, they showed the optimality of cyclic skimming pricing strategy and identified conditions on parameters such that a high-low pricing strategy is optimal. Chen et al. (2016) studied a finite horizon dynamic pricing problem where demand in each period was assumed to be deterministic in a memory-based reference price model. Because the problem with asymmetric effects of reference price is non-smooth and no standard optimization methods can be applied, they developed a strongly polynomial time algorithm to compute the optimal prices for a special case of gain-seeking customers. Their algorithms were also applied in a case study with real data to demonstrate the efficiency and robustness.

Along with pricing, firms simultaneously focus on their inventory management to better align supply with demand. Lack of information about relevant inventory costs can significantly affect the pricing decisions. Gimpl-Heersink et al. (2008) incorporated the inventory holding cost into a multiple-period inventory problem with reference price effects when customers are loss-neutral and demand is a linear function with an additive random term. They also demonstrated that a base-stock policy is optimal by assuming joint concavity of the demand and revenue functions. Because Gimpl-Heersink et al. (2008) showed the optimality of a base-stock policy under very strict assumptions, Chen et al. (2014) extended the model of Gimpl-Heersink et al. (2008) to focus on the loss-averse case by relaxing the assumption of joint concavity of the demand and revenue functions

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in a stochastic multi-period setting. In addition, they also analyzed their model over either a finite horizon or an infinite horizon. Güler et al. (2014) generalized the model of Gimpl-Heersink et al. (2008) to incorporate concave reference price effects and non-linear demand function (e.g., polynomial, exponential, and logarithmic base demand functions) with a finite planning horizon when customers are either loss-neutral or loss-averse. Lately, Güler et al. (2015) extended the problem by using safety stock as a decision variable and characterized the steady state solution for the infinite horizon problem. Dye and Yang (2016) investigated reference price effects in a joint dynamic pricing and preservation technology investment model for a deteriorating inventory system with time-and-price sensitive demand. Because they tried to optimize price and preservation technology investment over a finite planning horizon, they did not discuss the behavior of the long term equilibrium selling price. Xue et al. (2016) investigated the continuous-time dynamic optimal pricing strategy with reference price effects for deteriorating items. Assuming that customers are loss-neutral, they found in numerical studies that reference prices have a substantial impact on a firm's pricing strategy and profit.

Furthermore, marketing researchers and practitioners have recognized the phenomenon that high inventories might stimulate demand in years because large amounts of inventory might signal a popular product, or provide consumers with an assurance of high service levels. Hence, Gupta and Vrat (1986) extended the economic ordering quantity model (EOQ) for the case where demand rate depends on initial inventory. Baker and Urban (1988) assumed that demand rate is a power function of the current inventory level. Datta and Pal (1990) modified the model of Baker and Urban (1988) by assuming that the stock-dependent demand rate is down to a given level of inventory, beyond which it is a constant. Furthermore, deterioration is a common phenomenon in daily life because of poor storage and preservation quality. Padmanabhan and Vrat (1995), Dye and Ouyang (2005) and Min and Zhou (2009) developed deteriorating inventory models with linear stock-dependent demand and partial backlogging rates by using the known selling price. Chang (2004) maximized the profit instead of minimizing the costs in an inventory system for deteriorating items with an inventory-level-dependent demand rate. She also relaxed the terminal condition of zero ending-inventory to allows for a positive ending-inventory. Balakrishnan et al. (2004) demonstrated that the profit-maximizing policy orders larger quantities and yields significantly higher profits than cost-based inventory policies when inventories stimulate demand. Later, Yang et al. (2010) discussed a partial backlogging inventory lot-sizing model under inflation for deteriorating items with linear stock-dependent demand rate and used the discounted total profit as their objective.

Price is often the main factor in the purchasing decision of consumers even though demand is stock dependent. Additionally, it is an effective lever to manage the profitability in retailing, especially for the ubiquity of deteriorating items in our daily life. Urban and Baker (1997) formulated asingle periodmodel by incorporating the effects of price, time, and inventory level. Datta and Paul (2001) developed a finite period system with stock-dependent and price-sensitive demand rate and considered positive stock level just before placing an order. Teng and Chang (2005) extended an economic production quantity model (or EPQ) for perishable items, in which the demand rate depends on both the stock level and the selling price. Hou and Lin (2006) developed a finite planning horizon inventory model for deterioration items with price- and stock-dependent selling rates and shortages under inflation. You and Hsieh (2007) considered the demand rate is linearly dependent on selling price and stock level when the firm divides a given sales season into n periods. Even under the model without deterioration, they pointed out that it is difficult to analyze the optimal decisions for general value of n and therefore they only discussed three special cases $n = 1, 2, 3$. Lee and Dye

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(2012) addressed the inventory management problems by incorporating stock- and price-dependent demand and preservation technology investment. They also indicated that the retailer can reduce the economic losses, improve the customer service level and increase business competitiveness with the efforts of investing in reducing the deterioration rate. Saha et al. (2012) and Panda (2013) both studied the coordination of supply chain with stock- and price-dependent selling rates. Recently, Lu et al. (2016) established a finite planning horizon optimal control model for deteriorating item to seek the optimal joint dynamic pricing and replenishment strategy, where the demand rate is dependent on the on-hand stocks and the sales price. In addition, they also observed from their numerical results that the joint dynamic pricing and replenishment policy is better than the joint static pricing and dynamic replenishment policy. However, the above-mentioned studies assumed that the demand depends only on current price and displayed inventory level, and ignored the internal reference price mechanisms developed by customers on the basis of past observed prices for the product.

Finally, we summarize the major assumptions and objectives used in the above research articles in Table 1 to make it easier for the readers to understand the contributions of our model.

Table 1: Summary of the related research

Authors	deterministic/stochastic demand	reference price	price-dependent demand	stock-dependent demand	Deterioration
Kopalle and Winer (1996)	Deterministic	Yes	Linear	No	No
Urban and Baker (1997)	Deterministic	No	Isoelastic	Yes	No
Datta and Paul (2001)	Deterministic	No	General Form	Yes	No
Fibich et al. (2003)	Deterministic	Yes	Linear	No	No
Chang (2004)	Deterministic	No	No	Yes	Yes
Teng and Chang (2005)	Deterministic	No	General Form	Yes	Yes
Hou and Lin (2006)	Deterministic	No	General Form	Yes	Yes
Popescu and Wu (2007)	Deterministic	Yes	General form	No	No
You and Hsieh (2007)	Deterministic	No	Linear	Yes	No
Gimpl-Heersink et al. (2008)	Stochastic	Yes	Linear	No	No
Min and Zhou (2009)	Deterministic	No	No	Yes	Yes
Yang et al. (2010)	Deterministic	No	No	Yes	Yes
Nasiry and Popescu (2011)	Deterministic	Yes	General form	No	No
Saha et al. (2012)	Deterministic	No	Linear	Yes	No
Panda (2013)	Deterministic	No	Linear	Yes	No
Chen et al. (2014)	Stochastic	Yes	Linear	No	No
Güler et al. (2014)	Stochastic	Yes	General form	No	No
Xue et al. (2016)	Deterministic	Yes	Linear	No	Yes
Yang (2014)	Deterministic	No	No	Yes	No
Güler et al. (2015)	Stochastic	Yes	General form	No	No
Dye and Yang (2016)	Deterministic	Yes	General form	No	Yes
Hu et al. (2016)	Deterministic	Yes	Linear	No	No
Chen et al. (2016)	Deterministic	Yes	Linear	No	No
Present paper	Deterministic	Yes	General form	Yes	Yes

3. Model formulation

Consider a retailer who maintains an inventory of a single deteriorating product to meet a deterministic demand. Shortages are not allowed to avoid lost sales. Suppose ordering cycles are identical and each cycle is of equal length T in an infinite planning horizon. During the in-stock period, the product deteriorates at a constant rate θ, $0 < \theta \ll 1$, and there is no repair or replacement of deteriorated units during the replenishment period. Note that Goyal and Giri (2001) classified the deteriorating inventory models into the following three categories: (1) models for inventory with a fixed lifetime, (2) models for inventory with a random lifetime, and (3) models

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for inventory which decays corresponding to the proportional inventory decrease in terms of its utility or physical quantity. Because we assumed that a constant proportion of physical quantity of items deteriorate over time, our model belongs to the third type of this classification, and the deterioration in this paper means the physical inventory losses.

Furthermore, because the displayed stock level may bring about increased sales and most customers compare the current and previously observed prices of products to adjust their purchases, we assume that market demand depends not only the on-display stock level but also the selling price and the reference price. Let p_i and r_i denote the selling price and reference price, respectively, during the ith replenishment period, where $\underline{p} \leq p_i \leq \overline{p}$ and $\underline{p} \leq r_i \leq \overline{p}$ for $i = 1, 2, 3, \ldots$. At time t in the ith replenishment period, we assume that customers arrive at a rate of $G(p_i, r_i, t)$. For simplicity, we assume the additive demand function including reference price effects in the following way:

$$G(p_i, r_i, t) \equiv D(p_i, r_i) + \beta I(t),$$

where β is a non-negative constant, $I(t)$ is the stock on display at time t, and

$$\begin{aligned} D(p_i, r_i) &\equiv d(p_i) - \lambda \times \max\{0, p_i - r_i\} - \gamma \times \min\{0, p_i - r_i\} \\ &= \begin{cases} d(p_i) - \lambda\,(p_i - r_i)\,, & \text{if } p_i > r_i, \\ d(p_i) & \text{if } p_i = r_i, \\ d(p_i) - \gamma\,(p_i - r_i)\,, & \text{if } p_i < r_i. \end{cases} \end{aligned} \tag{1}$$

Note that the first term in (1) represents a standard demand model with generalized price effects. The use of two reference price parameters, λ and γ, create an asymmetric effect on the market demand $G(p_i, r_i, t)$ about r_i during the ith replenishment period when $\lambda \neq \gamma$. Since the reference price is the price that consumers anticipate to pay for a particular good or service, r_i can be seen as a market reference price for the deteriorating product. Furthermore, λ is a parameter corresponding to how the consumers view a reference price that is less than the observed price, and γ is a parameter corresponding to how the consumers view a reference price that is greater than the observed price. Therefore, the reference price parameters, λ and γ, can be seen as average loss averseness in the market. It is straightforward to see that if the observed price p_i is lower than the reference price r_i during the ith replenishment period, there is a positive gap between the reference price and the observed price, and it would result in a perceived gain and customers are more attracted to make the purchase. On the contrary, there is a negative gap between the reference price and the observed price if the observed price is greater than the reference price, and it would result in a perceived loss and customers are less attracted to make the purchase. Note that the market demand in this model is affected by the market-level average reference price, rather than the behaviors of individual customers. In order to formulate the problem more generalized, $d(p_i)$ is assumed to be non-negative, bounded, continuously differentiable, decreasing and convex in selling price. Furthermore, we assume that $D(p_i, r_i)$ is positive for all $p_i \in [\underline{p}, \overline{p}]$, and both λ and γ are positive to ensure $D(p_i, r_i)$ is decreasing in selling price and non-decreasing in reference price. If $\lambda = \gamma$, market demand is loss-neutral because demand responds equally to the effect of perceived gains and losses. For the loss-averse market demand, we let $\lambda > \gamma$ and demand is more responsive to perceived losses than equally sized gains. If $\lambda < \gamma$, market demand is loss-seeking.

Moreover, reference price is based on the existing reference price and observed prices in each replenishment period. We adopt a commonly used model, exponential smoothing, for the evolution of the reference price. Following Greenleaf (1995), Kopalle and Winer (1996), Fibich et al. (2003)

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and Popescu and Wu (2007), the reference price r_i during the ith replenishment period is defined as follows:

$$r_i = \alpha r_{i-1} + (1-\alpha)p_{i-1},$$

where α is the memory parameter and $0 \leq \alpha \leq 1$. Historical prices strongly affect the future reference price if α is high. Conversely, a low α implies that consumers have a short-term memory and the recent selling price has a greater effect on the reference price. Because customers have no information about the retailer's costs, we assume the initial reference price, r_1, at the beginning of the planning horizon is known and constant. Consequently, the reference price during the ith replenishment period can be expressed as follows:

$$r_i = \alpha r_{i-1} + (1-\alpha)\, p_{i-1} = \alpha^{i-1} r_1 + (1-\alpha) \sum_{m=1}^{i-1} \alpha^{i-1-m} p_m,\ i = 2, 3, 4, \ldots \tag{2}$$

Finally, we introduce other notation used throughout this paper. Let c ($c < \underline{p}$) be the unit purchasing cost and h be the holding cost per unit per unit time. Without loss of generality, we assume that the delivery lead time is zero and orders are delivered immediately every nT period, where n is a positive integer. The inventory level at time t is $I(t)$. We set a maximum inventory level w to reflect the facts that most retail outlets have limited shelf space. At time $t = 0$, we assume the initial stock in hand is given as $E_0 = 0$, and the first replenishment is made. With the progress of time, a fresh replenishment arrives at a time when the stock level reduces to the ending inventory level $E_i \geq 0$, $i = 1, 2, 3, \ldots$.

Base on the above assumptions and notation, the depletion of the inventory occurs because of the combined effects of demand and deterioration during the ith replenishment period, and we can write the inventory dynamics by using the following differential equation:

$$\begin{aligned} \frac{\mathrm{d}I(t)}{\mathrm{d}t} &= -G(p_i, r_i, t) - \theta I(t) \\ &= -D(p_i, r_i) - (\beta + \theta)\, I(t),\ (i-1)nT < t < inT, \end{aligned} \tag{3}$$

with the boundary condition $I(inT) = E_i$. Solving the differential equation (3) yields

$$I(t) = \frac{D(p_i, r_i)}{\beta + \theta} \left[e^{(\beta+\theta)(inT - t)} - 1 \right] + E_i e^{(\beta+\theta)(inT-t)},\ (i-1)nT \leq t \leq inT. \tag{4}$$

From (4), the initial inventory on hand and the time-weighted inventory during the ith replenishment period can be determined as

$$I((i-1)nT) = \frac{D(p_i, r_i)}{\beta + \theta} \left[e^{(\beta+\theta)nT} - 1 \right] + E_i e^{(\beta+\theta)nT} \tag{5}$$

and

$$\int_{(i-1)nT}^{inT} I(t)\mathrm{d}t = \frac{D(p_i, r_i)}{(\beta+\theta)^2} \left[e^{(\beta+\theta)nT} - (\beta+\theta)\, nT - 1 \right] + E_i \frac{e^{(\beta+\theta)nT} - 1}{\beta + \theta}, \tag{6}$$

respectively. Furthermore, the ordering quantity is $I((i-1)nT) - E_{i-1}$ and the amount of items sold during the ith replenishment period is

$$\int_{(i-1)nT}^{inT} \left[D(p_i, r_i) + \beta I\,(t) \right] \mathrm{d}t = D(p_i, r_i) \left\{ nT + \frac{\beta}{(\beta+\theta)^2} \left[e^{(\beta+\theta)nT} - (\beta+\theta)\, nT - 1 \right] \right\}$$

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$$+\frac{\beta E_i}{\beta+\theta}\left[e^{(\beta+\theta)nT}-1\right]. \tag{7}$$

In conjunction with the relevant costs mentioned above, the single period total profit during the ith replenishment period, denoted by $\pi(n,p_i,r_i,E_i)$, can be simplified as follows:

$$\begin{aligned}\pi(n,p_i,r_i,E_i) &= \text{sales revenues} - \text{purchase cost} - \text{holding cost}\\ &= (p_i-c)\,D(p_i,r_i)nT+[p_i\beta-h-c\,(\beta+\theta)]\,\frac{D(p_i,r_i)}{(\beta+\theta)^2}\left[e^{(\beta+\theta)nT}-(\beta+\theta)\,nT-1\right]\\ &\quad+\frac{E_i}{\beta+\theta}\left[e^{(\beta+\theta)nT}-1\right][p_i\beta-h-c\,(\beta+\theta)]-c\,(E_i-E_{i-1})\,. \end{aligned}\tag{8}$$

Let

$$\varphi(n,p)\equiv(p-c)\,nT+\frac{p\beta-h-c\,(\beta+\theta)}{(\beta+\theta)^2}\left[e^{(\beta+\theta)nT}-(\beta+\theta)\,nT-1\right], \tag{9}$$

denote the gross profit with zero ending inventory level during the ith replenishment period. It is straightforward to observe that $\pi(n,p_i,r_i,0)<0$ if $\varphi(n,p_i)<0$. Because the inventory system should not be operated if the gross profit is negative, we assume here that $\varphi(n,p)>0$ for $p\in[\underline{p},\overline{p}]$. Furthermore, because $\varphi(n,p)$ is increasing in p, setting (9) to zero yields the lower bound of p, denoted by p_l, as

$$p_l=\frac{c\left[e^{(\beta+\theta)nT}-1\right](\beta+\theta)+h\left[e^{(\beta+\theta)nT}-(\beta+\theta)\,nT-1\right]}{(\beta+\theta)\,\theta nT+\beta\left[e^{(\beta+\theta)nT}-1\right]}. \tag{10}$$

Without loss of generality, we make the following assumption on the lower bound of selling price $\underline{p}$ to avoid negative total profit during the ith replenishment period.

Assumption 1. $\underline{p}>p_l=\frac{c\left[e^{(\beta+\theta)nT}-1\right](\beta+\theta)+h\left[e^{(\beta+\theta)nT}-(\beta+\theta)nT-1\right]}{(\beta+\theta)\theta nT+\beta\left[e^{(\beta+\theta)nT}-1\right]}$.

In this paper, we take present value of money into account in finding the optimal inventory and pricing strategies over an infinite planning horizon. We model the retailer's long term profit optimization problem as follows:

$$\begin{aligned}\max_{(\mathbf{p},\mathbf{E})\in\mathbf{\Omega}}\ \Pi(n,\mathbf{p},\mathbf{E}) &\equiv \sum_{i=1}^{\infty}z(n)^{i-1}\pi(n,p_i,r_i,E_i)\\ &=\sum_{i=1}^{\infty}z(n)^{i-1}\varphi(n,p_i)D\left(p_i,\alpha^{i-1}r_1+(1-\alpha)\sum_{m=1}^{i-1}\alpha^{i-1-m}p_m\right)\\ &\quad+\sum_{i=1}^{\infty}z(n)^{i-1}E_i\left\{\frac{e^{(\beta+\theta)nT}-1}{\beta+\theta}[p_i\beta-h-c\,(\beta+\theta)]-[1-z(n)]c\right\}, \end{aligned}\tag{11}$$

where $z(n)\equiv\frac{1}{1+I_rnT}\in(0,1)$ is a discount factor, I_r is the interest rate per period, $\mathbf{p}=\{p_1,p_2,p_3,\dots\}$, $\mathbf{E}=\{E_1,E_2,E_3,\dots\}$ and

$$\mathbf{\Omega}=\left\{(n,\mathbf{p},\mathbf{E}):\underline{p}\le p_i\le\overline{p},\ E_i\ge0,\ \frac{D(p_i,r_i)}{\beta+\theta}\left[e^{(\beta+\theta)nT}-1\right]+E_ie^{(\beta+\theta)nT}\le w,\ i=1,2,3,\dots\right\}.$$

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By the assumption that $d(p)$ is convex in p, we can observe that the Hessian matrix of the constraint $\frac{D(p_i,r_i)}{\beta+\theta}\left[e^{(\beta+\theta)nT}-1\right]+E_i e^{(\beta+\theta)nT}-w$ is

$$\begin{bmatrix} d''(p_i) & 0 \\ 0 & 0 \end{bmatrix},$$

and so that

$$\begin{bmatrix} p_i & E_i \end{bmatrix} \begin{bmatrix} d''(p_i) & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_i \\ E_i \end{bmatrix} > 0$$

for all $\underline{p} \le p_i \le \overline{p}$ and $E_i \ge 0$, $i = 1, 2, 3, \ldots$, implying $\mathbf{\Omega}$ is a convex set for any given positive integer n.

Based the assumption of positive $\varphi(n,p)$, the following proposition characterizes the monotone behavior of the discounted total profit regarding the initial reference price.

Proposition 1. *The discounted total profit $\Pi(n,\mathbf{p},\mathbf{E})$ is an increasing function of the initial reference price r_1.*

From Proposition 1, one can have an intuition that the higher the initial reference price, the higher the future reference prices. This is because the exponential smoothing mechanism makes customers willing to pay higher prices, the discounted total profit increases as the initial reference price increases. Furthermore, based on the relationship between λ and γ in demand rate, there are three scenarios of market demand, namely loss-neutral, loss-averse and loss-seeking. We then derive the properties for these cases.

3.1. Market demand is loss-neutral

If market demand is loss-neutral (i.e., $\lambda = \gamma$), the problem in (11) can be then formulated as follows:

$$\begin{aligned}
\max_{(\mathbf{p},\mathbf{E})\in\mathbf{\Omega}} \Pi(n,\mathbf{p},\mathbf{E}) &\equiv \sum_{i=1}^{\infty} z(n)^{i-1}\pi(n,p_i,r_i,E_i) \\
&= \sum_{i=1}^{\infty} z(n)^{i-1}\varphi(n,p_i)\left\{d(p_i)-\lambda\left[p_i-\alpha^{i-1}r_1-(1-\alpha)\sum_{m=1}^{i-1}\alpha^{i-1-m}p_m\right]\right\} \\
&\quad + \sum_{i=1}^{\infty} z(n)^{i-1}E_i\left\{\frac{e^{(\beta+\theta)nT}-1}{\beta+\theta}\left[p_i\beta-h-c\left(\beta+\theta\right)\right]-\left[1-z(n)\right]c\right\}. \qquad (12)
\end{aligned}$$

Because $\Pi(n,\mathbf{p},\mathbf{E})$ is continuous and differentiable in $\mathbf{p}$ and $\mathbf{E}$ for a fixed n, the Hessian matrix of $\Pi(n,\mathbf{p},\mathbf{E})$ with respect to $\mathbf{p}$ and $\mathbf{E}$ can be expressed as

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{pp} & \mathbf{H}_{pE} \\ \mathbf{H}_{Ep} & \mathbf{H}_{EE} \end{bmatrix},$$

where

$$\mathbf{H}_{pp} = \begin{bmatrix} \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_1^2} & \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_1\partial p_2} & \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_1\partial p_3} & \cdots \\ \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_2\partial p_1} & \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_2^2} & \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_2\partial p_3} & \cdots \\ \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_3\partial p_1} & \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_3\partial p_2} & \frac{\partial^2\Pi(n,\mathbf{p},\mathbf{E})}{\partial p_3^2} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

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$$\mathbf{H}_{pE} = \text{diag}\left(\frac{\beta\left[e^{(\beta+\theta)nT}-1\right]}{\beta+\theta}, \frac{\beta\left[e^{(\beta+\theta)nT}-1\right]z}{\beta+\theta}, \frac{\beta\left[e^{(\beta+\theta)nT}-1\right]z(n)^2}{\beta+\theta}, \ldots\right)$$

and $\mathbf{H}_{EE}$ is a zero square matrix.

Let the ith term in the sequences $\{p_i^*\}_{i=1}^{\infty}$, $\{r_i^*\}_{i=1}^{\infty}$ and $\{E_i^*\}_{i=1}^{\infty}$, respectively, denote the optimal selling price, reference price and stock level just before placing an order during the ith replenishment period. The following proposition proves the strict concavity of $\Pi(n, \mathbf{p}, \mathbf{E})$ with respect to $\mathbf{p}$ under somewhat weaker conditions.

Lemma 1. *If $(p-c)\,d''(p)+d'(p) \leq 0$ and $d(p)$ is convex in p, then for loss-neutral market demand (i.e., $\lambda = \gamma$), $\Pi(n, \mathbf{p}, \mathbf{E})$ is strictly concave in $\mathbf{p}$.*

From Lemma 1, we directly have the following result.

Proposition 2. *Suppose that $(p-c)\,d''(p) + d'(p) \leq 0$ and $d(p)$ is convex in p. For loss-neutral market demand, if $\Pi(n, \mathbf{p}, \mathbf{E})$ attains its maximum value on $\mathbf{\Omega}$, then it is reached at some boundary point for any given n.*

Proposition 2 indicates that the optimal solution is always located on the boundary of $\mathbf{\Omega}$ because of the terms in the third line of (12). Note that the assumption $(p-c)\,d''(p) + d'(p) \leq 0$ implies that the gross profit without reference price effect $(p-c)d(p)$ is strictly concave in p. Furthermore, by rearranging the terms in the last bracket of (12), we note that $\beta(p_i - c)$ is the gross revenue received from an additional unit of inventory and $h + \theta c$ is the cost due to an additional unit of inventory. Hence, $\frac{e^{(\beta+\theta)nT}-1}{\beta+\theta}\left[p_i\beta - h - c\,(\beta+\theta)\right] \geq [1-z(n)]c$ represents the marginal gross profit gain from investing in an additional unit of ending inventory level is greater than or equal to the discount lost by not investing the ending inventory level within the discount period. Clearly, if $\frac{e^{(\beta+\theta)nT}-1}{\beta+\theta}\left[p_i\beta - h - c\,(\beta+\theta)\right] - [1-z(n)]c < 0$, i.e., $p_i < \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$, the lower E_i causes a larger value of $\Pi(n, \mathbf{p}, \mathbf{E})$, i.e., $E_i = 0$ for some i. On the other hand, if $p_i \geq \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$, it straightforward to see that it would be more profitable to hold greater inventory levels throughout the cycle, i.e., $E_i > 0$ for some i.

Because we have seen in Proposition 1 that the discounted total profit increases as the initial reference price increases, it is useful to understand the impacts of the initial reference price on the pricing policy. We next investigate how the initial reference price affects the optimal pricing decision, which is introduced in the following formal proposition.

Proposition 3. *If market demand is loss-neutral, $\{p_i^*\}_{i=1}^{\infty}$ increases as r_1 increases.*

Proposition 3 indicates that the optimal selling price and initial reference price are strategic complements. This result is intuitive because a higher reference price encourages customers to pay a higher selling price for the product than they would, the retailer can charge a higher selling price and hold more inventory level. Using this result, we prove the monotonicity behaviors of selling price and ending inventory level in the following proposition.

Proposition 4. *If market demand is loss-neutral, then $\{p_i^*\}_{i=1}^{\infty}$ is monotonic in the same direction as $\{r_i^*\}_{i=1}^{\infty}$. Furthermore, if there exists a $j \geq 2$ such that (i) $p_{j-1} < \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$*

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and $p_j > \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$ *or (ii)* $p_{j-1} > \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$ *and* $p_j < \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$*, then* $\{E_i\}_{i=j}^{\infty}$ *is also monotonic in the same direction as* $\{r_i^*\}_{i=j}^{\infty}$*.*

Proposition 4 reveals that $\{p_i^*\}_{i=1}^{\infty}$ and $\{r_i^*\}_{i=1}^{\infty}$ are both monotonic for any given initial reference price. The relationship between the monotonicity of optimal selling price and initial reference price can be interpreted as follows. When the initial reference price is relatively high, the retailer enters a market with a skimming strategy to target profits rather than market share. The retailer takes advantage of the consumers who prefer a relatively high selling price for a product in order to be leaders in the market earlier on, and then decreases the selling price gradually until the long term equilibrium selling price occurs. Therefore, the optimal selling price decreases over time when the initial reference price is relatively high. On the other hand, when the initial reference price is relatively low, the retailer enters a market with a penetration strategy to gain market share rapidly. The retailer sets a relatively low selling price for a product to attract customers earlier on, and then increases the selling price gradually to raise the future reference prices until the long term equilibrium selling price occurs. This explains why the optimal selling price path is increasing over the time when the initial reference price is relatively low.

For any given n, because the sequence $\{p_i^*\}_{i=1}^{\infty}$ is monotonic and bounded on the interval $[\underline{p}, \overline{p}]$, it converges to an equilibrium selling price as $i \to \infty$. Let $p_e^{\mathrm{LN}}(n)$ denote the long term equilibrium selling price for the case of loss-neutral market demand. Using the monotonic convergence of selling prices, the following proposition enables us to derive the upper bound of the optimal length of replenishment period for the problem in (12).

Proposition 5. *When market demand is loss-neutral, the optimal length of replenishment period must be less than or equal to* $\frac{1}{\beta+\theta}\ln\left[1+\frac{w(\beta+\theta)}{d(\overline{p})}\right]$*.*

Next, we consider two special inventory policies for the case of loss-neutral market demand as: (i) the zero ending inventory level policy, where $p_i < \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$ and $E_i = 0$, for all $i = 1, 2, 3, \ldots$, and (ii) the positive ending inventory level policy, where $p_i \geq \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$ and $E_i > 0$, for all $i = 1, 2, 3, \ldots$. For the zero ending inventory level policy, i.e., $\mathbf{E} = \mathbf{0}$, $\Pi(n, \mathbf{p}, \mathbf{E})$ becomes a function of n and $\mathbf{p}$. Let $\pi_{i1}^{\lambda}(n, p_i) \equiv \varphi(n, p_i) D(p_i, r_i)$ and $\Pi_1^{\lambda}(n, \mathbf{p})$, respectively, denote the single period total profit during the ith replenishment period and discounted total profit with the zero ending inventory level policy where $D(p, r) = d(p) - \lambda(p - r)$. The discounted total profit in (12) can be expressed as

$$\Pi_1^{\lambda}(n, \mathbf{p}) \equiv \sum_{i=1}^{\infty} z(n)^{i-1} \pi_{i1}^{\lambda}(n, p_i). \tag{13}$$

On the other hand, for the positive ending inventory level policy, i.e., $\mathbf{E} > \mathbf{0}$, the retailer tries to hold greater inventory levels. Therefore, it is straightforward to see that $\frac{D(p_i,r_i)}{\beta+\theta}\left[e^{(\beta+\theta)nT}-1\right] + E_i e^{(\beta+\theta)nT} = w$ from (5), i.e., $E_i = \frac{w}{e^{(\beta+\theta)nT}} - \frac{D(p_i,r_i)}{\beta+\theta}\left[1-\frac{1}{e^{(\beta+\theta)nT}}\right]$, $i = 1, 2, 3, \ldots$. Then we know that $\Pi(n, \mathbf{p}, \mathbf{E})$ becomes a function of n and $\mathbf{p}$. Let

$$\pi_{i2}^{\lambda}(n, p_i) \equiv \varphi(n, p_i) D(p_i, r_i) + \left\{\frac{w}{e^{(\beta+\theta)nT}} - \frac{D(p_i, r_i)}{\beta+\theta}\left[1 - \frac{1}{e^{(\beta+\theta)nT}}\right]\right\}$$
$$\times \left\{\frac{e^{(\beta+\theta)nT}-1}{\beta+\theta}\left[p_i\beta - h - c(\beta+\theta)\right] - [1 - z(n)]c\right\}$$

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and $\Pi_2^\lambda(n, \mathbf{p})$, respectively, denote the single period total profit during the ith replenishment period and discounted total profit with the positive ending inventory level policy where $D(p, r) = d(p) - \lambda(p - r)$. The discounted total profit in (12) can be rewritten as

$$\Pi_2^\lambda(n, \mathbf{p}) \equiv \sum_{i=1}^{\infty} z(n)^{i-1} \pi_{i2}^\lambda(n, p_i). \tag{14}$$

Let $p_{1e}^\lambda(n)$ and $p_{2e}^\lambda(n)$ denote, respectively, the long term equilibrium selling price for the inventory policy with $\mathbf{E} = \mathbf{0}$ and $\mathbf{E} > \mathbf{0}$. In order to characterize the equilibrium selling price for the two special inventory policies of the problem in (12) as mentioned above (i.e., $\mathbf{E} = \mathbf{0}$ and $\mathbf{E} > \mathbf{0}$), we need to establish the following lemmas.

Lemma 2. *Assuming an interior long term equilibrium selling price exists, for the inventory policy with zero ending inventory level (i.e., $\mathbf{E} = \mathbf{0}$), the long term equilibrium selling price $p_{1e}^\lambda(n)$ in the case of loss-neutral market demand must satisfy the equation:*

$$\begin{aligned} 0 = {} & \frac{d(p_{1e}^\lambda(n))}{1 - z(n)} \left\{ nT + \frac{\beta}{(\beta + \theta)^2} \left[e^{(\beta+\theta)nT} - (\beta + \theta) nT - 1 \right] \right\} \\ & + \varphi(n, p_{1e}^\lambda(n)) \left[\frac{d'(p_{1e}^\lambda(n))}{1 - z(n)} - \frac{\lambda}{1 - z(n)\alpha} \right]. \end{aligned} \tag{15}$$

Furthermore, $p_{1e}^\lambda(n)$ decreases strictly as λ increases.

Lemma 3. *Assuming an interior long term equilibrium selling price exists, for the inventory policy with positive ending inventory level (i.e., $\mathbf{E} > \mathbf{0}$), the long term equilibrium selling price $p_{2e}^\lambda(n)$ in the case of loss-neutral market demand must satisfy the equation:*

$$\begin{aligned} 0 = {} & \frac{d(p_{2e}^\lambda(n))}{1 - z(n)} \frac{\beta - \beta e^{-(\beta+\theta)nT} + (\beta + \theta)\theta nT}{(\beta + \theta)^2} \\ & + \frac{w\beta}{[1 - z(n)](\beta + \theta)} \left[1 - \frac{1}{e^{(\beta+\theta)nT}} \right] + \varphi(n, p_{2e}^\lambda(n)) \left[\frac{d'(p_{2e}^\lambda(n))}{1 - z(n)} - \frac{\lambda}{1 - z(n)\alpha} \right] \\ & - \frac{1}{\beta + \theta} \left[1 - \frac{1}{e^{(\beta+\theta)nT}} \right] \left\{ \frac{e^{(\beta+\theta)nT} - 1}{\beta + \theta} \left[p_{2e}^\lambda(n)\beta - h - c(\beta + \theta) \right] - [1 - z(n)]c \right\} \\ & \times \left[\frac{d'(p_{2e}^\lambda(n))}{1 - z(n)} - \frac{\lambda}{1 - z(n)\alpha} \right]. \end{aligned} \tag{16}$$

Furthermore, $p_{2e}^\lambda(n)$ decreases strictly as λ increases.

Lemmas 2 and 3 demonstrate how to find the long term equilibrium selling prices, p_{1e}^λ and p_{2e}^λ, and how the parameter λ affects the long term equilibrium selling prices for the inventory policies with $\mathbf{E} = \mathbf{0}$ and $\mathbf{E} > \mathbf{0}$ if an interior long term equilibrium selling price exists. This result is intuitive from (15) and (16). An increase in λ will lower the profit margin of the discounted total profit from the vth period to infinity (i.e., $\Pi_1^\lambda(n, p_{1e}^\lambda(n), p_{1e}^\lambda(n), \ldots | r_v)$ and $\Pi_2^\lambda(n, p_{2e}^\lambda(n), p_{2e}^\lambda(n), \ldots | r_v)$); and therefore the retailer should decrease its long term equilibrium selling price to avoid losing sales.

For any given positive integer n, let $p_e^{\mathrm{LN}}(n)$ and $E_e^{\mathrm{LN}}(n)$, respectively, denote the long term equilibrium selling price and long term equilibrium ending inventory level for the model when

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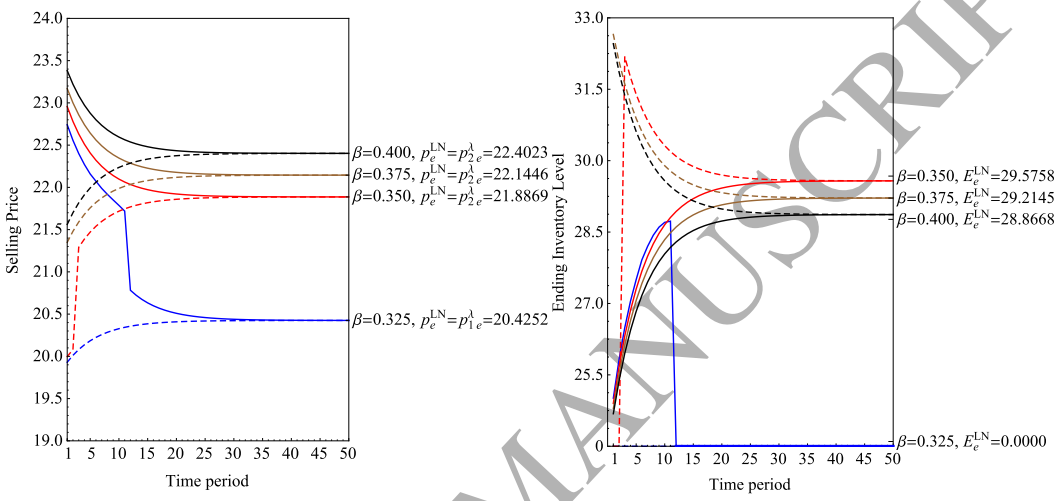
market demand is loss-neutral. With the help of the above propositions and lemmas, we can now give the long term equilibrium selling price and long term equilibrium ending inventory level for the model with loss-neutral market demand in the following theorem.

Theorem 1. *Assuming an interior long term equilibrium selling price exists in the case of loss-neutral market demand with a given n, $p_e^{\mathrm{LN}}(n)$ must converge to either $p_{1e}^{\lambda}(n)$ or $p_{2e}^{\lambda}(n)$, and $E_e^{\mathrm{LN}}(n)$ must converge to either 0 or $\frac{w}{e^{(\beta+\theta)nT}} - \frac{d(p_{1e}^{\lambda}(n))}{\beta+\theta}\left[1 - \frac{1}{e^{(\beta+\theta)nT}}\right]$*



(a) The optimal selling price (b) The optimal ending inventory level

Figure 1: The global optimal selling price and ending inventory level paths for the loss-neutral market demand ($\lambda = 2$, $\gamma = 2$). The blue, dashed blue, red, dashed red, brown, dashed brown, black and dashed black lines denote the global optimal selling price path with $(r_1, \beta) = (28, 0.325)$, $(17.6, 0.325)$, $(28, 0.35)$, $(17.6, 0.35)$, $(28, 0.375)$, $(17.6, 0.375)$, $(28, 0.4)$, $(17.6, 0.4)$, respectively.

We use Fig. 1 to demonstrate the numerical results of Propositions 3-4 and Theorem 1 for a parameter setting similar to Section 4 with $d(p) = 130 - 4p$, $n = 1$ and $w = 100$. Because of the supermodularity of $\Pi(n, \mathbf{p}, \mathbf{E})$ with respect to $(\mathbf{p}, \mathbf{E}, r_1)$ for all $p_i > \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$, one can observe from Fig. 1 that the monotonicity behaviors of selling price and ending inventory level. Fig. 1a shows that the optimal selling price evolutes over time until they reach the corresponding long term equilibrium selling price. Fig. 1b shows that retailer will hold greater inventory levels throughout the cycle such that $E_i > 0$ if it is beneficial for the retailer to invest in an additional unit of ending inventory level (i.e., $p_i > \frac{(\beta+\theta)[1-z(n)]c}{\beta\left[e^{(\beta+\theta)nT}-1\right]} + \frac{h+c(\beta+\theta)}{\beta}$), otherwise the retailer will hold zero inventory at the end of the cycle.

In the end of this subsection, we prove the uniqueness of the optimal pricing strategy for the special inventory policy with zero ending inventory level (i.e., $\mathbf{E} = \mathbf{0}$) in Theorem 2. Further, we also reveal the relationships between the initial reference price and optimal pricing strategy under the case of loss-neutral market demand. Proposition 6, as outlined below, characterizes the effect of initial reference price on the pricing path for the case of loss-neutral market demand. We omit the proofs as they follow immediately from Lemmas 1–3 and Proposition 4.

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Theorem 2. *For the inventory policy with zero ending inventory level, if* $(p-c)\,d''(p)+d'(p)\leq 0$, $d(p)$ *is convex in* p *and* n *is gven, then a unique* $\mathbf{p}$ *exists such that* $\Pi_1^{\lambda}(n,\mathbf{p})$ *is maximized when market demand is loss-neutral (i.e.,* $\lambda=\gamma$*).*

Proposition 6. *Assuming an interior long term equilibrium selling price exists for a given* n*. For the inventory policy with zero ending inventory level (i.e.,* $\mathbf{E}=\mathbf{0}$*), when market demand is loss-neutral, we have:*

1. *If* $r_1>p_{1e}^{\lambda}(n)$, $\{p_i^*\}_{i=1}^{\infty}$ *and* $\{r_i^*\}_{i=1}^{\infty}$ *both decrease monotonically.*
2. *If* $r_1<p_{1e}^{\lambda}(n)$, $\{p_i^*\}_{i=1}^{\infty}$ *and* $\{r_i^*\}_{i=1}^{\infty}$ *both increase monotonically.*
3. *If* $r_1=p_{1e}^{\lambda}(n)$, $r_1=p_1^*=p_2^*=p_3^*=\cdots=p_{\infty}^*$ *and* $r_1=r_2^*=r_3^*=\cdots=r_{\infty}^*$.

3.2. Market demand is loss-averse

If market demand is loss-averse (i.e., $\lambda>\gamma$), the problem in (11) can be then formulated as follows:

$$\max_{(\mathbf{p},\mathbf{E})\in\mathbf{\Omega}}\ \Pi(n,\mathbf{p},\mathbf{E})=\sum_{i=1}^{\infty}z(n)^{i-1}\varphi(n,p_i)\Big\{d(p_i)-\lambda\max\{0,p_i-r_i\}-\gamma\min\{0,p_i-r_i\}\Big\}$$
$$+\sum_{i=1}^{\infty}z(n)^{i-1}E_i\left\{\frac{e^{(\beta+\theta)nT}-1}{\beta+\theta}\left[p_i\beta-h-c\left(\beta+\theta\right)\right]-\left[1-z(n)\right]c\right\}. \quad (17)$$

Because the demand is kinked at $p_i=r_i=\alpha^{i-1}r_1+(1-\alpha)\sum_{m=1}^{i-1}\alpha^{i-1-m}p_m$, $\Pi(n,\mathbf{p},\mathbf{E})$ is continuous but not differentiable everywhere for a fixed n. Consequently, the first order necessary condition for the optimality is no longer valid in this circumstance. Let $\pi^{\lambda}(n,p,r,E)$ and $\pi^{\gamma}(n,p,r,E)$ respectively denote the single period total profit with $D(p,r)=d(p)-\lambda\left(p-r\right)$ and $D(p,r)=d(p)-\gamma\left(p-r\right)$. To solve the problem for this case, we need the following lemma.

Lemma 4. *If* $\lambda\geq\gamma>0$ *and* $p\geq p_l$ *as (10), then* $\pi^{\lambda}(n,p,r,E)\geq\pi(n,p,r,E)$ *and* $\pi^{\gamma}(n,p,r,E)\geq\pi(n,p,r,E)$.

Let $\Pi_{\text{LN}}^{\lambda}(n,\mathbf{p},\mathbf{E})$ and $\Pi_{\text{LN}}^{\gamma}(n,\mathbf{p},\mathbf{E})$ respectively denote the discounted total profit for the problem (11) with $D(p,r)=d(p)-\lambda\left(p-r\right)$ and $D(p,r)=d(p)-\gamma\left(p-r\right)$. With the help of Lemma 4, it is easy to show that $\Pi(n,\mathbf{p},\mathbf{E})\leq\Pi_{\text{LN}}^{\lambda}(n,\mathbf{p},\mathbf{E})$ and $\Pi(n,\mathbf{p},\mathbf{E})\leq\Pi_{\text{LN}}^{\gamma}(n,\mathbf{p},\mathbf{E})$, which is equivalent to $\Pi(n,\mathbf{p},\mathbf{E})\leq\min\{\Pi_{\text{LN}}^{\lambda}(n,\mathbf{p},\mathbf{E}),\Pi_{\text{LN}}^{\gamma}(n,\mathbf{p},\mathbf{E})\}$. Therefore, the problem in (17) can be rewritten as

$$\max_{(\mathbf{p},\mathbf{E})\in\mathbf{\Omega}}\ \Pi(n,\mathbf{p},\mathbf{E})=\min\{\max_{(\mathbf{p},\mathbf{E})\in\mathbf{\Omega}}\Pi_{\text{LN}}^{\lambda}(n,\mathbf{p},\mathbf{E}),\max_{(\mathbf{p},\mathbf{E})\in\mathbf{\Omega}}\Pi_{\text{LN}}^{\gamma}(n,\mathbf{p},\mathbf{E})\}, \quad (18)$$

Our next result shows the supermodularity of $\Pi(n,\mathbf{p},\mathbf{E})$ with respect to $(\mathbf{p},\mathbf{E},r_1)$ and the monotonicity property of $\{p_i^*\}_{i=1}^{\infty}$, $\{E_i^*\}_{i=1}^{\infty}$ and $\{r_i^*\}_{i=1}^{\infty}$ for the case of loss-averse market demand. Because the proofs follow immediately from Propositions 3 and 4, we omit the proofs to respect the space limitations.

Proposition 7. *If market demand is loss-averse,* $\{p_i^*\}_{i=1}^{\infty}$ *and* $\{E_i^*\}_{i=1}^{\infty}$ *both increases as* r_1 *increases. Furthermore,* $\{p_i^*\}_{i=1}^{\infty}$ *and* $\{E_i^*\}_{i=1}^{\infty}$ *are monotonic in the same direction as the reference price path.*

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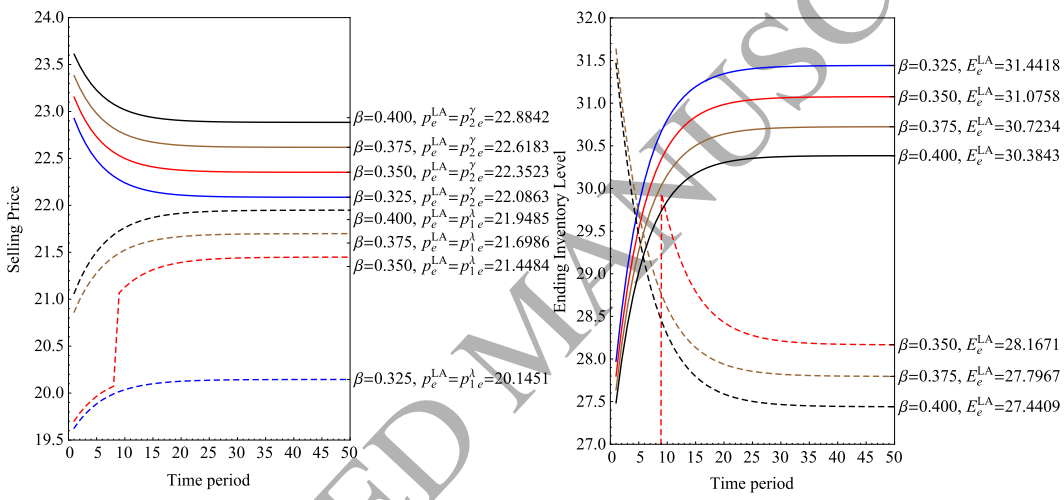
For any given n, because the sequence $\{p_i^*\}_{i=1}^{\infty}$ is monotonic and bounded on the interval $[\underline{p}, \overline{p}]$, it converges to an equilibrium selling price as $i \to \infty$. For any given positive integer n, let $p_e^{\mathrm{LA}}(n)$ and $E_e^{\mathrm{LA}}(n)$, respectively, denote the long term equilibrium selling price and long term equilibrium ending inventory level for the model when market demand is loss-averse. We also let $p_{1e}^{\gamma}(n)$ and $p_{2e}^{\gamma}(n)$ denote, respectively, the long term equilibrium selling price for the inventory policy with $\mathbf{E} = \mathbf{0}$ and $\mathbf{E} > \mathbf{0}$ when demand function $D(p, r) = d(p) - \gamma(p - r)$. With the help of the above propositions and lemmas, we can characterize the long term equilibrium selling price for the problem in (17) in the following theorem.

Theorem 3. *Assuming an interior long term equilibrium selling price exists in the case of loss-averse market demand with a given n, the long term equilibrium selling price for the problem in (17), $p_e^{\mathrm{LA}}(n)$, must converge to r_1, $p_{1e}^{\lambda}(n)$, $p_{2e}^{\lambda}(n)$, $p_{1e}^{\gamma}(n)$ or $p_{2e}^{\gamma}(n)$.*



(a) The optimal selling price (b) The optimal ending inventory level

Figure 2: The global optimal selling price and ending inventory level paths for the loss-averse market demand ($\lambda = 2.5$, $\gamma = 1.5$). The blue, dashed blue, red, dashed red, brown, dashed brown, black, and dashed black lines denote the global optimal selling price path with $(r_1, \beta) = (28, 0.325)$, $(17.6, 0.325)$, $(28, 0.35)$, $(17.6, 0.35)$, $(28, 0.375)$, $(17.6, 0.375)$, $(28, 0.4)$, and $(17.6, 0.4)$, respectively.

Theorem 3 reveals that the long term equilibrium selling price for the inventory system depends on the initial reference price, and Fig. 2 demonstrates its numerical results for a parameter setting similar to Section 4 with $d(p) = 130 - 4p$, $n = 1$ and $w = 100$. If the initial reference price $r_1 > p_{1e}^{\gamma}$ ($r_1 > p_{2e}^{\gamma}$), the optimal selling price path decreases monotonically and converges to the long term equilibrium selling price p_{1e}^{γ} (p_{2e}^{γ}). On the other hand, if the initial reference price $r_1 < p_{1e}^{\lambda}$ ($r_1 < p_{2e}^{\lambda}$), the optimal selling price path increases monotonically and converges to its long term equilibrium selling price p_{1e}^{λ} (p_{2e}^{λ}). Note that if the initial reference price r_1 lying between p_{1e}^{λ} and p_{1e}^{γ} (p_{2e}^{λ} and p_{2e}^{γ}), the optimal values of decision variables $\{p_i\}_{i=1}^{\infty}$ equal to r_1, and thus the long term equilibrium selling price is already reached.

Recall that the optimal selling price and reference price would converge to the same equilibrium point as $i \to \infty$, it is clearly to to see that $p_i - r_i = 0$ as $i \to \infty$. Furthermore, we have the

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following analogue of Proposition 5 immediately, which shows the upper bound of the optimal length of replenishment period for the problem in (17).

Proposition 8. *When market demand is loss-averse, the optimal length of replenishment period must be less than or equal to* $\frac{1}{\beta+\theta}\ln\left[1+\frac{w(\beta+\theta)}{d(\overline{p})}\right]$.

In the end of this subsection, we let $\Pi_{\text{LA}}(n,\mathbf{p})$ denote the discounted total profit with the zero ending inventory level policy (i.e., $\mathbf{E}=\mathbf{0}$) for the case of loss-averse market demand. We also let $\Pi^{\lambda}_{\text{LN}}(n,\mathbf{p})$ and $\Pi^{\gamma}_{\text{LN}}(n,\mathbf{p})$ respectively denote the discounted total profit with $D(p,r)=d(p)-\lambda(p-r)$ and $D(p,r)=d(p)-\gamma(p-r)$ when market demand is loss-neutral and $\mathbf{E}=\mathbf{0}$. We then prove the uniqueness of the optimal pricing strategy for the special inventory policy in the Theorem 4. Further, we also reveal the relationships between the initial reference price and optimal pricing strategy under the case of loss-averse market demand. Proposition 9, as outlined below, characterizes the effect of initial reference price on the pricing path for the case of loss-averse market demand. We omit the proofs as they follow immediately from Lemma 4, Proposition 7 and Theorem 3.

Theorem 4. *Assuming an interior long term equilibrium selling price exists for a given n. For the inventory policy with zero ending inventory level, if $(p-c)\,d''(p)+d'(p)\le 0$ and $d(p)$ is convex in p, then a unique $\mathbf{p}$ exists such that $\Pi_{\text{LA}}(n,\mathbf{p})$ is maximized for the case of loss-averse market demand (i.e., $\lambda>\gamma$).*

Proposition 9. *Assuming an interior long term equilibrium selling price exists for a fixed n. For the inventory policy with zero ending inventory level (i.e., $\mathbf{E}=\mathbf{0}$), when market demand is loss-averse, we have:*

1. *If $r_1>p^{\gamma}_{1e}(n)$, $\{p^*_i\}_{i=1}^{\infty}$ and $\{r^*_i\}_{i=1}^{\infty}$ both decrease monotonically to the long term equilibrium selling price $p^{\gamma}_{1e}(n)$, and $\max_{\mathbf{p}\in\mathbf{\Omega}}\Pi_{\text{LA}}(n,\mathbf{p})=\max_{\mathbf{p}\in\mathbf{\Omega}}\Pi^{\gamma}_{\text{LN}}(n,\mathbf{p})$.*
2. *If $r_1<p^{\lambda}_{1e}(n)$, $\{p^*_i\}_{i=1}^{\infty}$ and $\{r^*_i\}_{i=1}^{\infty}$ both increase monotonically to the long term equilibrium selling price $p^{\lambda}_{1e}(n)$, and $\max_{\mathbf{p}\in\mathbf{\Omega}}\Pi_{\text{LA}}(n,\mathbf{p})=\max_{\mathbf{p}\in\mathbf{\Omega}}\Pi^{\lambda}_{\text{LN}}(n,\mathbf{p})$.*
3. *If $p^{\lambda}_{1e}(n)\le r_1\le p^{\gamma}_{1e}(n)$, $r_1=p^*_1=p^*_2=p^*_3=\cdots=p^*_{\infty}$.*

3.3. Market demand is loss-seeking

If market demand is loss-seeking (i.e., $\gamma>\lambda$), the problem in (11) can be then formulated as follows:

$$\begin{aligned}\max_{(\mathbf{p},\mathbf{E})\in\mathbf{\Omega}}\Pi(n,\mathbf{p},\mathbf{E})&=\sum_{i=1}^{\infty}z(n)^{i-1}\max\left\{\varphi(n,p_i)\left[d(p_i)-\lambda\left(p_i-r_i\right)\right],\varphi(n,p_i)\left[d(p_i)-\gamma\left(p_i-r_i\right)\right]\right\}\\&\quad+\sum_{i=1}^{\infty}z(n)^{i-1}E_i\left\{\frac{e^{(\beta+\theta)nT}-1}{\beta+\theta}\left[p_i\beta-h-c\left(\beta+\theta\right)\right]-\left[1-z(n)\right]c\right\}\\&=\sum_{i=1}^{\infty}z(n)^{i-1}\max\{\pi^{\lambda}(n,p_i,r_i,E_i),\pi^{\gamma}(n,p_i,r_i,E_i)\}.\end{aligned}\tag{19}$$

Clearly, the discounted total profit $\Pi(n,\mathbf{p},\mathbf{E})$ is still continuous but not differentiable everywhere for a fixed n as the reason in Subsection 3.2. To solve the problem for this case, we need the following lemma.

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Lemma 5. *If $0 < \lambda < \gamma$ and $\underline{p} \geq p_l$ as (10), then $\pi^{\lambda}(n,p,r,E) > \pi^{\gamma}(n,p,r,E)$ if $p > r$ and $\pi^{\lambda}(n,p,r,E) \leq \pi^{\gamma}(n,p,r,E)$ if $p \leq r$.*

We now denote $\boldsymbol{\delta} = \{\delta_1, \delta_2, \delta_3, \dots\}$ and let

$$\delta_i = \begin{cases} 1, \, p_i > r_i = \alpha^{i-1} r_1 + (1-\alpha) \sum_{m=1}^{i-1} \alpha^{i-1-m} p_m, \\ 0, \, p_i \leq r_i = \alpha^{i-1} r_1 + (1-\alpha) \sum_{m=1}^{i-1} \alpha^{i-1-m} p_m, \end{cases}$$

$i = 1, 2, 3, \dots$. Note that $\boldsymbol{\delta}$ can be considered pricing alternative that the retailer selects over the planning horizon; and therefore, the problem in (19) can be equivalently written as the optimization problem:

$$\max_{\boldsymbol{\delta} \in \{0,1\}^{\infty}} \max_{\mathbf{p} \in \boldsymbol{\Omega}_{\text{LS}}} \Gamma(n, \mathbf{p}, \mathbf{E}, \boldsymbol{\delta}),$$

where

$$\Gamma(n, \mathbf{p}, \mathbf{E}, \boldsymbol{\delta}) \equiv \sum_{i=1}^{\infty} z(n)^{i-1} \left\{ \delta_i \times \pi^{\lambda}(n, p_i, r_i, E_i) + (1-\delta_i) \times \pi^{\gamma}(n, p_i, r_i, E_i) \right\} \tag{20}$$

and

$$\boldsymbol{\Omega}_{\text{LS}} = \boldsymbol{\Omega} \cap \left\{ \mathbf{p} : \begin{cases} p_i > r_i = \alpha^{i-1} r_1 + (1-\alpha) \sum_{m=1}^{i-1} \alpha^{i-1-m} p_m, \, \delta_i = 1, \\ p_i \leq r_i = \alpha^{i-1} r_1 + (1-\alpha) \sum_{m=1}^{i-1} \alpha^{i-1-m} p_m, \, \delta_i = 0, \end{cases}, \, i = 1, 2, 3, \dots \right\}.$$

Note also that $\boldsymbol{\Omega}_{\text{LS}}$ is a convex set. In this section, as similar as in Sections 3.1 and 3.2, we cannot show the strict concavity of $\Gamma(n, \mathbf{p}, \mathbf{E}, \boldsymbol{\delta})$ in $\mathbf{p}$ and $\mathbf{E}$. Even the strict concavity of $\Gamma(n, \mathbf{p}, \mathbf{E}, \boldsymbol{\delta})$ in $\mathbf{p}$ and $\mathbf{E}$ for any given n and $\boldsymbol{\delta}$, the maximum of concave functions is not concave generally. Therefore, in the case of loss-seeking market demand the profit function $\Pi(n, \mathbf{p}, \mathbf{E})$ is not concave, and we cannot prove rigorously the uniqueness of the optimal value of $\mathbf{p}$ and $\mathbf{E}$ as n is fixed.

From (19), it is easy to show that $\Pi(n, \mathbf{p}, \mathbf{E}) \geq \Pi^{\lambda}_{\text{LN}}(n, \mathbf{p}, \mathbf{E})$ and $\Pi(n, \mathbf{p}, \mathbf{E}) \geq \Pi^{\gamma}_{\text{LN}}(n, \mathbf{p}, \mathbf{E})$ from (19), which is equivalent to $\Pi(n, \mathbf{p}, \mathbf{E}) \geq \max\{\Pi^{\lambda}_{\text{LN}}(n, \mathbf{p}, \mathbf{E}), \Pi^{\gamma}_{\text{LN}}(n, \mathbf{p}, \mathbf{E})\}$. Therefore, we can obtain the lower bound of the problem in (19). With the help of Lemma 5, furthermore, it is straightforward to see that $\Gamma(n, \mathbf{p}, \mathbf{E}, \boldsymbol{\delta}) > \Pi^{\lambda}_{\text{LN}}(n, \mathbf{p}, \mathbf{E})$ for all $\boldsymbol{\delta} \neq \mathbf{1}$ and $\Gamma(n, \mathbf{p}, \mathbf{E}, \boldsymbol{\delta}) > \Pi^{\lambda}_{\text{LN}}(n, \mathbf{p}, \mathbf{E})$ for all $\boldsymbol{\delta} \neq \mathbf{0}$. As opposed to Propositions 3, 4 and 7, which show that $\Pi(n, \mathbf{p}, \mathbf{E})$ is supermodular in $(\mathbf{p}, \mathbf{E}, r_1)$ for loss-neutral and loss-averse market demand, we are unable to prove the same result for $\Pi(n, \mathbf{p}, \mathbf{E})$ with respect to $(\mathbf{p}, \mathbf{E}, r_1)$ for loss-seeking market demand. If market demand is more responsive to gains, however, we can show that the optimal pricing strategy cycles, namely a High-Low pricing strategy. In the case, the retailer charges the item at a higher price to establish the price image in the minds of consumers and then use price discounts and promotions frequently for a limited period of time to attract price-sensitive customers. This result can be stated in the following formal lemma.

Lemma 6. *The optimal pricing strategy for the case of loss-seeking market demand is High-Low pricing.*

Furthermore, we have the following analogue of Proposition 5 immediately, which shows the upper bound of the optimal length of replenishment period for the problem in (19).

Proposition 10. *When market demand is loss-seeking, the optimal length of replenishment period must be less than or equal to $\frac{1}{\beta+\theta} \ln \left[1 + \frac{w(\beta+\theta)}{d(\overline{p})} \right]$.*

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In the end of this subsection, we let $\Pi_{\mathrm{LS}}(n, \mathbf{p})$ denote the discounted total profit with the zero ending inventory level policy (i.e., $\mathbf{E} = \mathbf{0}$) for the case of loss-seeking market demand. The discounted total profit in (19) can be rewritten as

$$\Pi_{\mathrm{LS}}(n, \mathbf{p}) = \max_{\boldsymbol{\delta} \in \{0,1\}^{\infty}} \Gamma_1(n, \mathbf{p}, \boldsymbol{\delta}),$$

where

$$\Gamma_1(n, \mathbf{p}, \boldsymbol{\delta}) \equiv \sum_{i=1}^{\infty} z(n)^{i-1} \varphi(n, p_i) \left[d(p_i) + \lambda (r_i - p_i) \delta_i + \gamma (r_i - p_i)(1 - \delta_i) \right]. \tag{21}$$

Therefore, the special inventory policy of the problem in (19) as $\mathbf{E} = \mathbf{0}$ can be equivalently written as the optimization problem:

$$\max_{\boldsymbol{\delta} \in \{0,1\}^{\infty}} \max_{\mathbf{p} \in \boldsymbol{\Omega}_{\mathrm{LS}}} \Gamma_1(n, \mathbf{p}, \boldsymbol{\delta}).$$

The following theorem reveals the uniqueness of the optimal $\mathbf{p}$ to $\Gamma_1(n, \mathbf{p}, \boldsymbol{\delta})$ for any given n and $\boldsymbol{\delta}$. We also show that the optimal pricing strategy must be High-Low pricing in the following formal propositions. We omit the proofs as they follow immediately from the supermodularity of $\Gamma(n, \mathbf{p}, \mathbf{E}, \boldsymbol{\delta})$ with respect to $(\mathbf{p}, \mathbf{E}, r_1)$ as n and $\boldsymbol{\delta}$ are fixed.

Theorem 5. *For the inventory policy with zero ending inventory level, if $(p - c)\, d''(p) + d'(p) \leq 0$, $d(p)$ is convex in p, then for loss-seeking market demand (i.e., $\lambda < \gamma$), a unique $\mathbf{p}$ exists such that $\Gamma_1(n, \mathbf{p}, \boldsymbol{\delta})$ is maximized as n and $\boldsymbol{\delta}$ are fixed.*

Proposition 11. *For the inventory policy with zero ending inventory level, the optimal pricing strategy for the case of loss-seeking market demand is High-Low pricing.*

4. Numerical example and sensitivity analysis

In this section we consider groups of numerical examples to verify the obtained analytical results. The base settings of parameters are listed as follows: $\underline{p} = 15$, $\overline{p} = 30$, $T = 1$, $c = 10$, $h = 2$, $\alpha = 0.8$, $I_r = 0.25$, $\theta = 0.1$ and $w = 250$. The solution procedure is implemented using Mathematica Version 9 on a personal computer with an Intel Core i5 processor and running Mac OS X 10.9.2. By using the results from Propositions 5, 8 and 10, we can obtain that the upper bound of n is 8 with $d(p) = 130 - 4p$ and $\beta = 0.1$. Table 2 presents the optimal discounted total profit, the long term equilibrium selling price and the long term equilibrium ending inventory level with respect to different values of n (from 1 to 8) for the three scenarios of market demand, and Fig. 3 illustrates the corresponding results of the optimal pricing paths.

For the scenarios of loss-neutral and loss-averse market demand in Fig. 3a and 3b, we can observe that the optimal selling price evolutes over time until it reaches the long term equilibrium selling price as indicated in Propositions 4 and 7. If the initial reference price is high, the retailer adopts a skimming strategy to target profits earlier on, i.e., it starts with a high selling price and gradually decreases it until the long term equilibrium selling price occurs. On the other hand, if the initial reference price is low, the retailer adopts a penetration strategy to gain market share rapidly, i.e., it starts with a low selling price and gradually increases it until the long term equilibrium selling price occurs.

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Table 2: The optimal discounted total profit with respect to different values of n for the three scenarios of market demand, with $d(p) = 130 - 4p$.

Reference price	Loss-neutral $\lambda = \gamma = 2$				Loss-averse $\lambda = 2.5$ and $\gamma = 1.5$				Loss-seeking $\lambda = 1.5$ and $\gamma = 2.5$	
r_1	n	$\Pi(n, p^*, E^*)$	p_e^{LN}	E_e^{LN}	n	$\Pi(n, p^*, E^*)$	p_e^{LA}	E_e^{LA}	n	$\Pi(n, p^*, E^*)$
17.6	1	2118.28	20.7323	0	1	2084.77	20.4599	0	1	2157.41
17.6	2	2178.92	21.3234	0	2	2123.75	21.0229	0	2	2243.67
17.6	3	2140.36	22.1242	0	3	2060.29	21.8298	0	3	2233.66
17.6	4	2004.37	23.0045	0	4	1897.76	22.7274	0	4	2128.14
17.6	5	1781.26	25.2229	0	5	1648.81	25.2206	0	5	1935.36
17.6	6	1471.14	27.1096	0	6	1322.85	27.1064	0	6	1644.05
17.6	7	1121.45	28.4054	0	7	967.869	28.4018	0	7	1300.83
17.6	8	779.012	29.3346	0	8	625.006	29.3306	0	8	959.141
28	1	2695.88	20.7323	0	1	2599.61	21.0228	0	1	2793.02
28	2	3018.61	21.3234	0	2	2891.64	21.6485	0	2	3146.67
28	3	3167.53	22.1242	0	3	3030.79	22.4447	0	3	3305.72
28	4	3170.96	23.0045	0	4	3042.04	23.3072	0	4	3284.77
28	5	2895.17	25.2229	0	5	2838.96	25.2260	0	5	2943.52
28	6	2477.47	27.1096	0	6	2461.34	27.1131	0	6	2491.31
28	7	2044.13	28.4054	0	7	2038.33	28.4072	0	7	2050.93
28	8	1635.03	29.3346	0	8	1617.49	29.3364	0	8	1655.57

For the scenario of loss-seeking market demand with $\lambda = 1.5$ and $\gamma = 2.5$, Fig. 3c shows that the optimal selling prices tend to be increasing with $r_1 = 17.6$ and to be decreasing with $r_1 = 28$ earlier on. After several periods, the optimal selling price path is formed with a cyclic High-Low price pattern which was indicated in Lemma 6. The result can be interpreted as follows. When the initial reference price is relatively low, because selling price reductions cause a sustained decrease in the price image that is not beneficial in the long run, the retailer should first increase the product selling price frequently to create high reference prices. Until the reference price reaches acceptable levels, since the customers weigh gains more heavily than equally sized losses, the retailer repeatedly increases selling prices to raise the reference price and reduces selling prices to earn more profits. On the other hand, when the initial reference price is relatively high, because increasing selling price results in the difficulty in market development in the long run, the retailer first decreases the selling price frequently to attract more customers who perceive they are getting a favorable price. Before the selling price falls to the level unfavorable to the retailer's profitability of sales promotion, the retailer increases the selling price over the reference price to improve profitability and to avoid a negative impression from a low level price. Then the retailer adjusts the selling prices among different high and low levels repeatedly because customers are more sensitive to gains than losses.

Furthermore, we can also observe from Fig. 4 that the optimal selling prices increase as n increases for the three scenarios of market demand. Note that the initial inventory level in (5) increases as the length of replenishment period, i.e., nT, increases. A higher value of n leading to the higher initial inventory level not only induces more demand but also helps the retailer generate more marginal profit gains from an additional unit of inventory. Therefore, the retailer should increase the selling price to earn more profits.

Next, we perform some numerical studies to further investigate the influences of the initial reference price r_1, the memory factor α, the shape parameter β, deterioration rate θ and per unit

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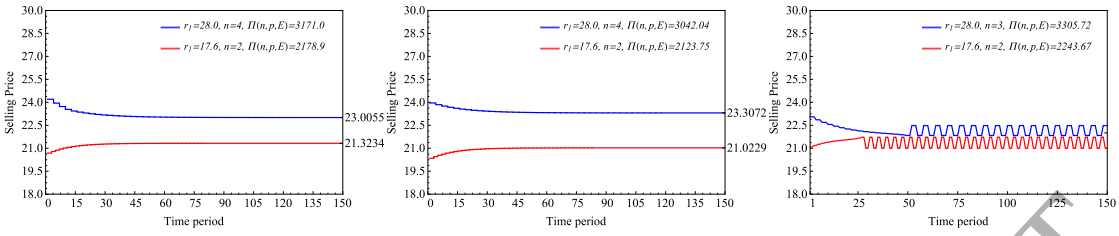

(a) $\lambda = 2$, $\gamma = 2$ (b) $\lambda = 2.5$, $\gamma = 1.5$ (c) $\lambda = 1.5$, $\gamma = 2.5$

Figure 3: Optimal selling price paths for the three scenarios: loss-neutral, loss-averse and loss-seeking market demand, with $d(p) = 130 - 4p$.

holding cost h on the retailer's optimal discounted total profit and replenishment strategy. All the computed results of these sensitive analysis are shown in Table 3. The numerical results first illustrate that a higher initial reference price results in a higher discounted total profit as Proposition 1 indicates. A potential reason is that a higher value of r_1 leads to higher reference prices in the future through the exponential smoothing mechanism, motivating customers to pay a higher price for the product, and then increases the optimal discounted total profit. Table 3 also illustrates that a higher initial reference price results in a longer replenishment period. The intuition is as following. When the initial reference price is low, the retailer would raise its selling price frequently to increase the reference price because a higher reference price stimulates buying. However, when the initial reference price is high, the retailer targets its discounted total profit rather than market share and thus reduces the replenishment frequency to maintain the higher reference price of the product.

Moreover, Table 3 reveals that the discounted total profit increases with an increase in the memory factor α for the three scenarios of market demand. This happens because a higher value of α represents customers are strongly impressed on the past reference prices, the retailer can continue offering price promotions to stimulate sales when the initial reference price is high, leading to an increase in the discounted total profit. Meanwhile, because the retailer adopts a penetration strategy to gain market share rapidly when the initial reference price is low, the increasing selling price causes the marginal gross profit received from an additional unit of inventory, i.e., $\beta p - h - c(\beta + \theta)$ increases. Hence, we can also find that the discounted total profit for the case of loss-neutral or loss-averse market demand increases as α increases. However, we observe that the discounted total profit for the case of loss-seeking market demand decreases with an increase in the memory factor α when r_1 is low. This is because it is not beneficial for the retailer to implement a High-Low pricing strategy with a higher long-term memory of the historical prices of a product, the discounted total profit decreases with α for the case of loss-seeking market demand when the initial reference price is low.

Likewise, we also observe that the optimal discounted total profit and length of replenishment period both increase as the sharp parameter β increases. When β increases, it represents that the marginal gross profit received from an additional unit of inventory increases. Hence, the optimal discounted total profit increases if β increases. Meanwhile, because the initial inventory level increases as the length of replenishment period increases and the inventories stimulate the demand rate, the optimal length of replenishment period increases as β increases. Finally, it is straight-

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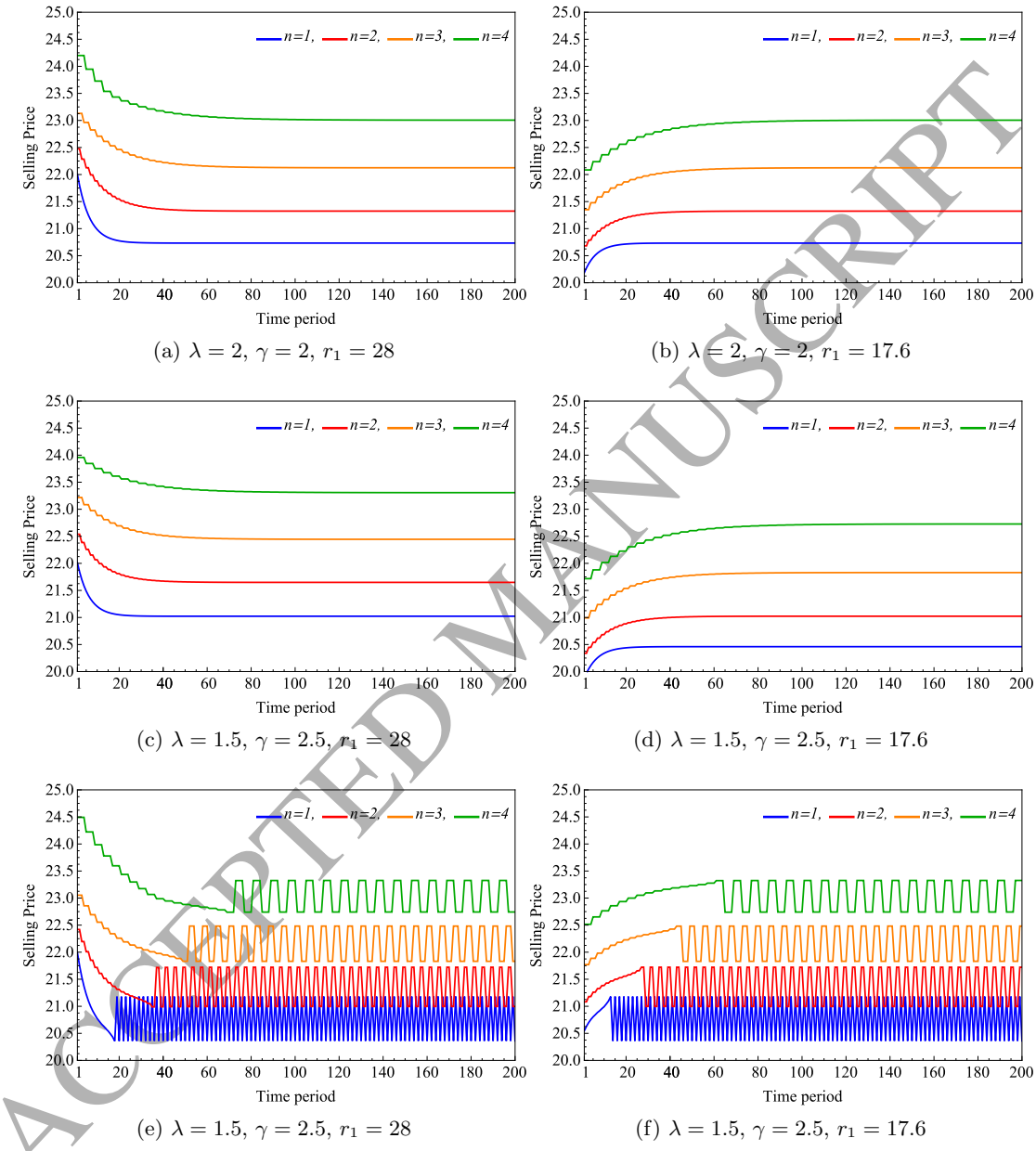

(a) $\lambda = 2$, $\gamma = 2$, $r_1 = 28$

(b) $\lambda = 2$, $\gamma = 2$, $r_1 = 17.6$

(c) $\lambda = 1.5$, $\gamma = 2.5$, $r_1 = 28$

(d) $\lambda = 1.5$, $\gamma = 2.5$, $r_1 = 17.6$

(e) $\lambda = 1.5$, $\gamma = 2.5$, $r_1 = 28$

(f) $\lambda = 1.5$, $\gamma = 2.5$, $r_1 = 17.6$

Figure 4: The optimal selling price paths with respect to different values of n for the three scenarios of market demand, with $d(p) = 130 - 4p$.

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Table 3: Effects of changes in the parameters, with $d(p) = 130 - 4p$

Parameters				Loss-neutral $\lambda = \gamma = 2$		Loss-averse $\lambda = 2.5$ and $\gamma = 1.5$		Loss-seeking $\lambda = 1.5$ and $\gamma = 2.5$	
				n^*	Π^*	n^*	Π^*	n^*	Π^*
α	0.4	r_1	17.6	2	2172.7	2	2115.1	3	2333.7
	0.6		17.6	2	2176.8	2	2120.7	2	2294.4
	0.8		17.6	2	2178.9	2	2123.8	2	2243.7
	0.4		28	4	3169.8	4	3039.5	4	3141.9
	0.6		28	4	3170.7	4	3041.2	4	3205.4
	0.8		28	4	3171.0	4	3042.0	3	3305.7
β	0.05	r_1	17.6	1	2060.4	1	2027.6	2	2108.4
	0.10		17.6	2	2178.9	2	2123.8	2	2243.7
	0.20		17.6	3	2664.7	4	2655.5	7	3455.9
	0.05		28	3	2854.9	3	2732.4	3	2978.7
	0.10		28	3	3167.5	4	3042.0	3	3305.7
	0.20		28	3	3851.5	4	3992.9	7	6037.2
θ	0.05	r_1	17.6	4	2451.6	3	2357.3	4	2568.1
	0.10		17.6	2	2178.9	2	2123.8	2	2243.7
	0.20		17.6	1	1986.7	1	1950.7	1	2028.3
	0.05		28	4	3721.6	5	3569.8	4	3872.8
	0.10		28	4	3171.0	4	3042.0	3	3305.7
	0.20		28	2	2621.7	2	2516.4	2	2728.1
h	1.5	r_1	17.6	3	2393.0	3	2317.0	4	2495.8
	2.0		17.6	2	2178.9	2	2123.8	2	2243.7
	2.5		17.6	1	2058.1	1	2023.5	2	2104.0
	1.5		28	4	3604.9	4	3473.0	4	3718.8
	2.0		28	4	3171.0	4	3042.0	3	3305.7
	2.5		28	3	2875.4	3	2755.2	3	2996.9

forward to see from Table 3 that the optimal discounted total profit and length of replenishment period both decrease as θ or h increases because a higher value of θ or h causes a higher unit inventory cost.

Furthermore, we conduct additional experimentation to provide further insight into the problem where the stock-sensitive and price-sensitive patterns of demand function are extended to an isoelastic functional form, i.e., $G(p_i, r_i, t) = [d(p_i) - \lambda \times \max\{0, p_i - r_i\} - \gamma \times \min\{0, p_i - r_i\}] I(t)^{\beta}$. By assuming $d(p) = 18000p^{-1.75}$ and $\beta = 0.15$ we obtain that the upper bound of n is 2. Table 4 presents the optimal discounted total profit, the long term equilibrium selling price and the long term equilibrium ending inventory level with respect to $n = 1, 2$ for the three scenarios of market demand; and Fig. 5 illustrates the corresponding results of the optimal selling price and ending inventory level paths. We also conduct sensitivity analyses to identify effects of parameters such as r_1, α, β, θ and h on the retailer's optimal discounted total profit and replenishment strategy, which are shown in Table 5. We can also observe from the numerical experimentation that there is no significant difference in the conclusions derived from the uses of different stock-sensitive and price-sensitive patterns of demand functions; and thus, the experimental results show the robustness of our proposed model.

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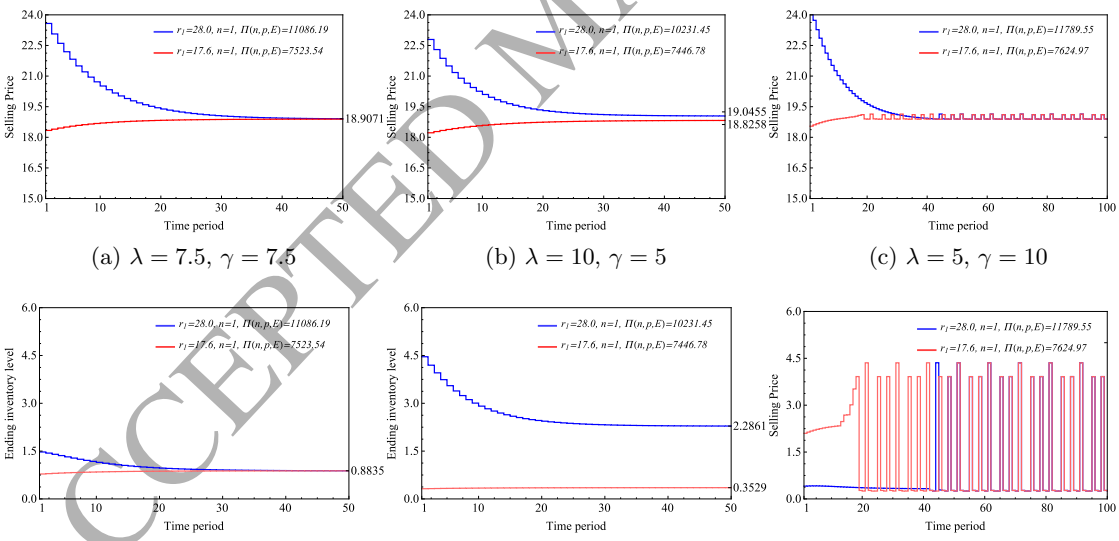
Table 4: The optimal discounted total profit with respect to different values of n for the three scenarios of market demand, with $d(p) = 18000p^{-1.75}$.

Reference price	Loss-neutral $\lambda = \gamma = 7.5$				Loss-averse $\lambda = 10$ and $\gamma = 5$				Loss-seeking $\lambda = 5$ and $\gamma = 10$	
r_1	n	$\Pi(n, p^*, E^*)$	p_e^{LN}	E_e^{LN}	n	$\Pi(n, p^*, E^*)$	p_e^{LA}	E_e^{LA}	n	$\Pi(n, p^*, E^*)$
17.6	1	7523.54	18.8912	0.8808	1	7446.78	18.8258	0.3529	1	7624.97
17.6	2	6307.18	27.6501	0.0000	2	6847.89	27.8155	0.0000	2	6847.93
28	1	11086.20	18.9071	0.8835	1	10231.45	19.0455	2.2861	1	11789.55
28	2	9674.63	27.9067	0.0001	2	9668.75	27.9048	0.0000	2	9678.20



(a) $\lambda = 7.5$, $\gamma = 7.5$ (b) $\lambda = 10$, $\gamma = 5$ (c) $\lambda = 5$, $\gamma = 10$

(d) $\lambda = 7.5$, $\gamma = 7.5$ (e) $\lambda = 10$, $\gamma = 5$ (f) $\lambda = 5$, $\gamma = 10$

Figure 5: Optimal selling price and ending inventory level paths for the three scenarios: loss-neutral, loss-averse and loss-seeking market demand, with $d(p) = 18000p^{-1.75}$.

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Table 5: Effects of changes in the parameters, with $d(p) = 18000p^{-1.75}$

	Parameters			Loss-neutral $\lambda = \gamma = 2$		Loss-averse $\lambda = 2.5$ and $\gamma = 1.5$		Loss-seeking $\lambda = 1.5$ and $\gamma = 2.5$	
				n^*	Π^*	n^*	Π^*	n^*	Π^*
α	0.4	r_1	17.6	1	7508.7	1	7433.9	1	7797.2
	0.6		17.6	1	7519.5	1	7443.5	1	7714.4
	0.8		17.6	1	7523.5	1	7446.8	1	7625.0
	0.4		28	1	11051.3	1	10199.3	1	10444.1
	0.6		28	1	11075.1	1	10220.0	1	10961.1
	0.8		28	1	11086.2	1	10231.4	1	11789.5
β	0.05		17.6	2	4901.9	1	4804.8	2	5134.6
	0.10		17.6	1	5990.8	1	6004.8	1	6052.9
	0.20		17.6	1	9012.8	1	8782.1	1	9316.7
	0.05		28	2	7818.0	2	7425.9	2	8097.8
	0.10		28	1	8974.5	2	8612.1	1	9801.2
	0.20		28	1	13074.6	1	12409.3	1	13598.0
θ	0.05	r_1	17.6	1	7529.7	1	7450.0	1	7636.7
	0.10		17.6	1	7523.5	1	7446.8	1	7625.0
	0.20		17.6	1	7518.2	1	7444.2	1	7611.9
	0.05		28	1	11095.2	1	10250.3	1	11794.0
	0.10		28	1	11086.2	1	10231.4	1	11789.5
	0.20		28	1	11075.4	1	10206.9	1	11785.7
h	1.5	r_1	17.6	1	7755.9	1	7678.0	1	7860.1
	2.0		17.6	1	7523.5	1	7446.8	1	7625.0
	2.5		17.6	1	7292.2	1	7216.2	1	7390.7
	1.5		28	1	11319.2	1	10469.8	1	12021.0
	2.0		28	1	11086.2	1	10231.4	1	11789.5
	2.5		28	1	10853.1	1	9994.1	1	11558.7

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5. Concluding Remarks

For deteriorating items having a demand rate that depends on displayed stock level and selling price simultaneously, we analyze the dynamic pricing problem where consumer's purchase decisions are influenced by reference price effects over an infinite horizon. Traditional optimization method instead of dynamic programming technique is employed to solve the optimal pricing strategy. We first prove that the discounted total profit increases with the initial reference price predicted by the retailer at the start of the planning horizon, and then show that the optimal pricing strategy is always located at the boundary of the domain. To appropriately incorporate reference price effects into an inventory problem when inventories stimulate demand, we discuss the impact of building up inventory on the sales for the positive ending inventory level case including an ending inventory level of zero. The optimal selling pricing and inventory strategies are characterized for the scenarios of loss-neutral, loss-averse and loss-seeking scenario of loss-seeking market demand, respectively. Our theoretical results indicate that the optimal selling price and reference price paths are both demonstrated to converge monotonically to the long term equilibrium selling price when market demand is loss-neutral or loss-averse. It revealed that the retailer adopts a skimming strategy to target profits earlier on if the initial reference price is high, but adopts a penetration strategy to gain market share rapidly if the initial reference price is low. The results also showed that the value of the long term equilibrium selling price decreases with customers' response parameter to losses. So it is essential for a retailer to anticipate customers' behavioral parameters such as initial reference price and the loss response by marketing research. When market demand is loss-seeking, however, we only show that the optimal selling price path is not monotonic; and further, we demonstrate by numerical examples that optimal selling prices fluctuate frequently among different high and low levels, i.e., a cyclic High-Low price pattern. With a High-Low pricing strategy, the retailer would charge higher prices on an everyday basis, but then runs promotions frequently for a limited period of time to attract price-sensitive customers (e.g., Vons, Safeway, Kroger and Rewe).

Furthermore, numerical experimentation is also used to investigate the impacts of the parameters, such as memory factor, sharp parameter, holding cost and deterioration rate, on the optimal discounted total profit and pricing strategy. From our numerical results, we found that the effects of the selected parameters on the optimal discounted total profit are similar regardless of the three scenarios of market demand. The optimal discounted total profit increases as the memory factor increases first but decreases when the memory factor is relatively high. The results suggested that strength of the memory factor is important for a retailer to measure because a high value of the memory factor represents consumers have a longer-term memory of perceived gains under a high initial reference price or perceived losses under a low initial reference price. On the other hand, since the impact effect of shape parameter is to higher the marginal gross profit and the impact effect of holding cost and deterioration rate is to lower the marginal gross profit, it is not so surprising that the optimal discounted total profit increases as the shape parameter increases, but decreases as holding cost or deterioration rate increases. Therefore, if a retailer attempts to improve sales and profits, large piles of consumer goods displayed in the shelves is a simple merchandising tool. In addition, it is profitable for the retailer to invest in preservation technologies to reduce the inventory cost (including holding and deterioration costs).

For future work, some extensions are noteworthy. The update rule for the reference prices in this study was assumed to be linear, but we could include different memory-based behavioral reference prices, such as a peak-end rule, in the model. Furthermore, we introduced the deterioration rate

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of products into the presented model as an exogenous variable. However, since the inevitable phenomena of deterioration of products erodes grocery retailers' low profit margins, effective control of deterioration of products is increasingly crucial for the success in grocery retailing. An empirical study by Richards (2011) showed that reducing perishable waste by 20% would increase the total store profit by 33%. Hence it is worthwhile extending the model with effective investment in preservation technologies to reduce the deterioration loss. Meanwhile, the analysis in the proposed model was carried out in the simple setting of a deterministic and multiplicative demand form to reflect the demand-stimulating role of retail inventories. Because we assumed that a constant proportion of physical quantity of items deteriorate over time, the model belongs to the third type of Goyal and Giri's (2001) classification, and the deterioration in this paper means the continuous decay of inventory item and physical inventory losses. Hence, if the demand can not be predicted well, we can extend the proposed model to a stochastic optimal control problem with stochastic demand (e.g., Alshamrani (2013) and Pan and Li (2015)).

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